

WASP-12b

R-0387

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_230131 · created 2026-07-18T22:34:54+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459974.80324 +/- 0.00092 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1101 +/- 0.0085
Transit depth (Rp/Rs)^2	1.21 +/- 0.19 [%]
Orbital Inclination (inc)	81.03 +/- 2.05
Airmass coefficient 1 (a1)	1.207 +/- 0.01
Airmass coefficient 2 (a2)	-0.1238 +/- 0.0043
Scatter in the residuals of the lightcurve fit is	0.84 %
Best Comparison Star	None
Optimal Aperture	4.84
Optimal Annulus	10.51
Transit Duration (day)	0.1178 +/- 0.0057

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1101 $\pm$ 0.0085 vs 0.1178 (deviation 0.7 σ)
Duration fit vs published	0.1178 d vs 0.125 d (deviation 0.98 σ)
Mid-time O-C	-1574.9 min
Depth SNR	10.0
Residual scatter	0.84 %

Light curve

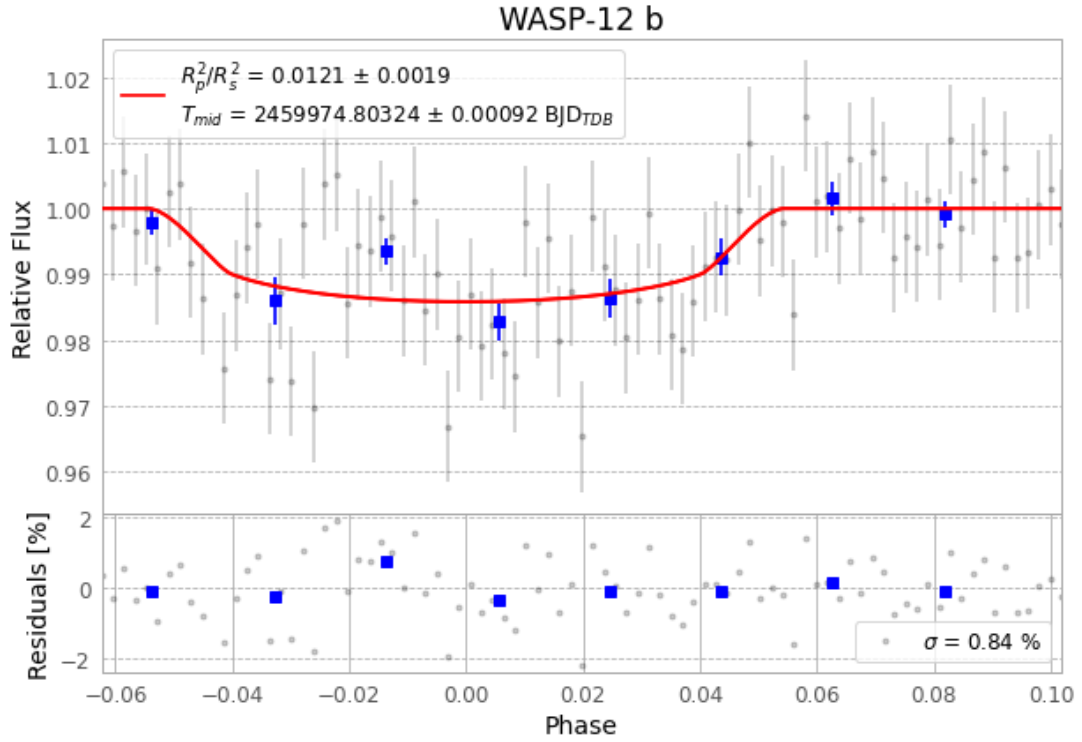


Figure 1 — FinalLightCurve_WASP-12 b_2023-01-29.png (SHA-256 68decd16359a6763...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [409, 220], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 147d13a13969647f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8350c6b

Data provenance

87 FITS frames (57 MB), WASP-12230130053015.FITS → WASP-12230130094815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-230131.log (46c4535eda4c...), FinalLightCurve_WASP-12 b_2023-01-29.png (68decd16359a...), FinalLightCurve_WASP-12 b_2023-01-29.csv (befda223c5f6...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 22:34Z.